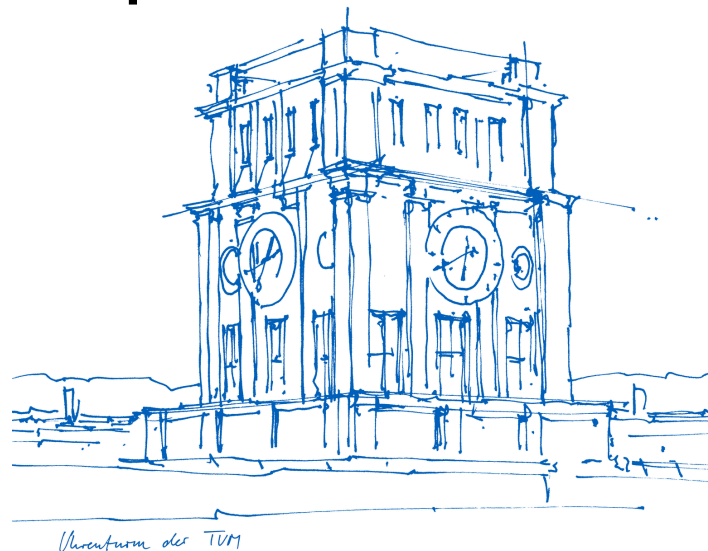


MUM2026, MATSim User Meeting

Measuring perturbation-reconvergence cost in MATSim for charging-infrastructure placement

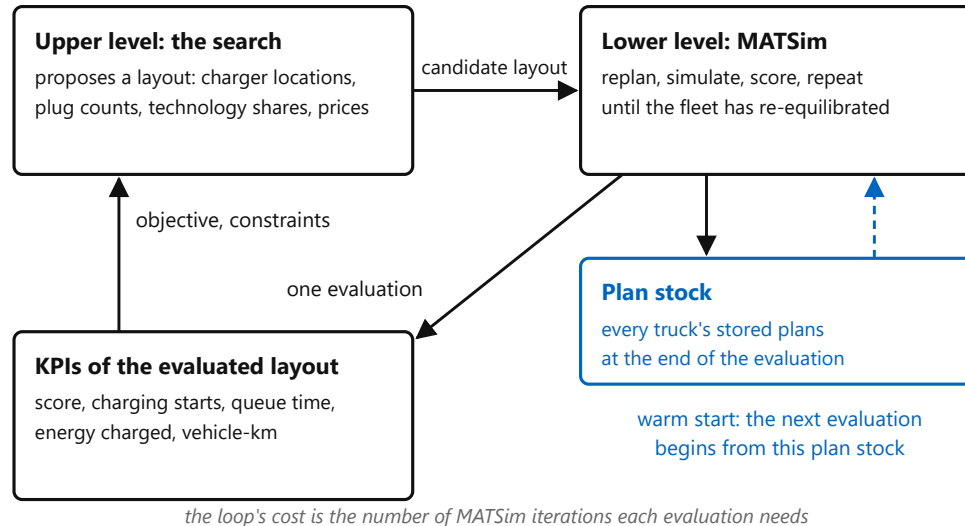
Shahriar Iqbal Zame
shahriar.zame@tum.de

Technical University of Munich
TUM School of Engineering and Design
Chair of Transportation Systems Engineering
Paris, 28 September 2026



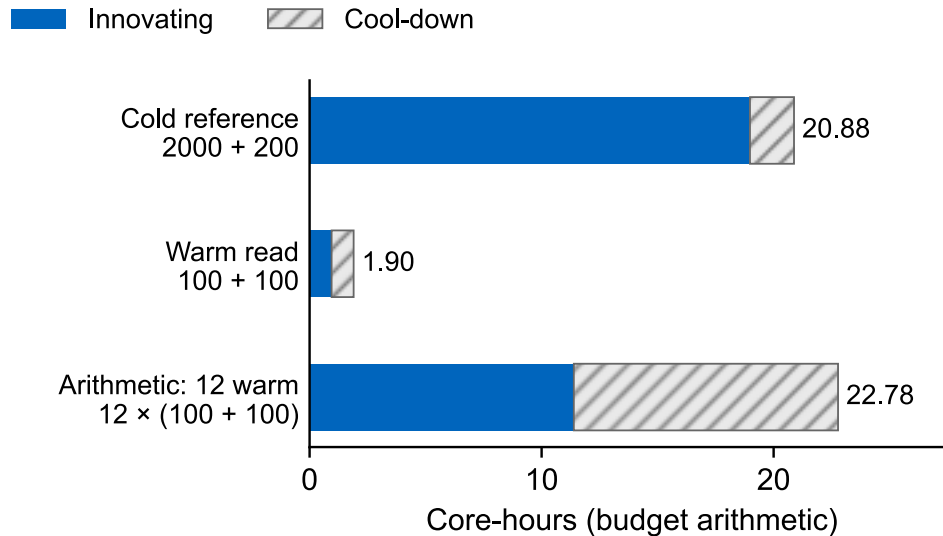
The idea and its price

An optimisation loop over a traffic simulation



- **Upper level:** a search over charger locations, plug counts, technology shares and prices
- **Lower level:** MATSim replans, simulates and scores a synthetic electric truck fleet until it has re-equilibrated
- **One search step costs one evaluation,** and every evaluation after the first can begin from the previous fleet's plans: a warm start

What one evaluation costs



6 seeds per candidate; common rate 5.69 s/iteration.
Warm / cold = 1/11; excludes parent setup.

Core-hours for a cold evaluation, a warm read and a twelve-candidate scenario, split into innovating and cool-down iterations

- One iteration is one simulated day: 250 trucks replan, drive and are scored. **5.5 to 6 s on one core.**
- A cold evaluation is **2000 innovating iterations plus a 200-iteration cool-down**: the reference budget
- A warm read at $k = 100$ with a 100-iteration cool-down is **200 of those 2200 iterations**, about one eleventh

How the field affords a simulation-based loop

#	Mechanism	Exemplars	What you need to know to use it well
1	Search smarter: fewer evaluations	Bayesian optimisation (Koutounidis et al., 2025); NSGA-II (Fadranski et al., 2026)	a design space and an evaluation budget
2	Run rougher: cheaper evaluations, same simulator	multi-fidelity methods (Peherstorfer et al., 2018); fixed 50-iteration inner budget (Patwary, Huang and Lo, 2021)	iterations to reconvergence against perturbation size: not reported
3	Carry state forward: reused evaluations	warm and hot starts (Nicolai, 2013); opdyts (Flötteröd, 2017)	iterations to reconvergence against perturbation size: not reported
4	Leave the simulator out: substituted evaluations	queueing surrogate (Fadranski et al., 2026); learned surrogates (Natterer et al., 2026)	a training set, or a simplified model you trust

We take row 3, and measure the number rows 2 and 3 both presuppose.

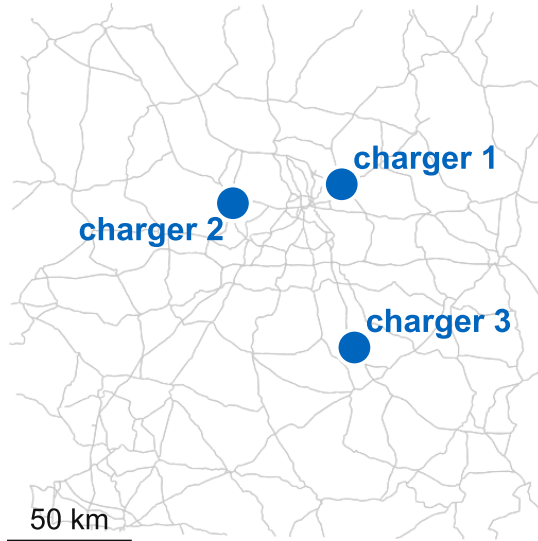
The instrument

What one iteration does

- One iteration is one simulated day: the trucks drive, and each executed plan is scored in utils
- Between days a truck either **picks one of its stored plans**, favouring better scores, or **rebuilds its route and charging stops from scratch**. Only the second creates anything new
- We call that second one **innovation**, and switching it off is the **shutoff**. **k** counts the innovating iterations a run is given

The scenario

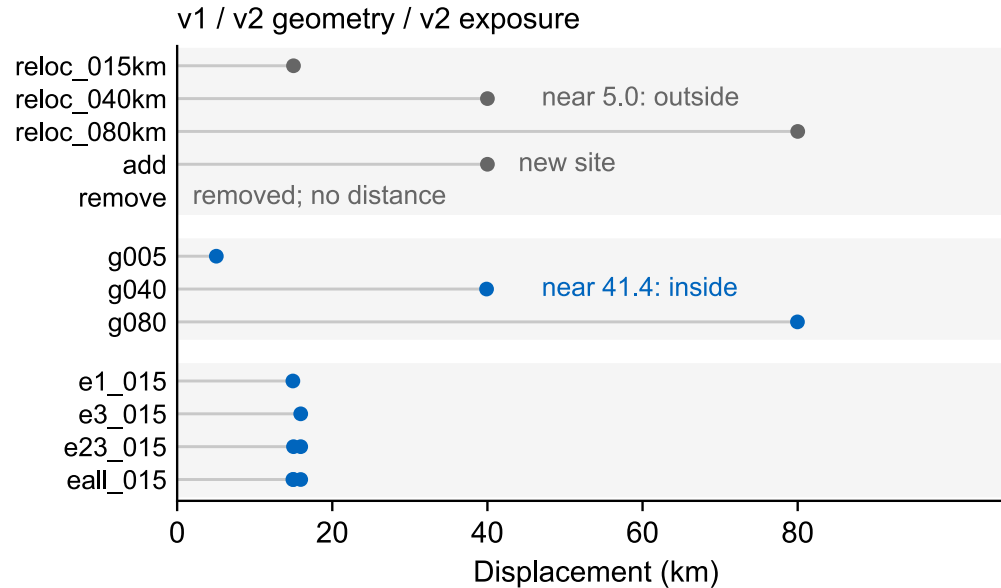
Brandenburg motorways, MATSim ev contrib; 250 trucks;
three 350 kW chargers, 8 plugs each



- Brandenburg motorways from the MATSim electric-vehicle contrib: 39,275 links, 24,432 nodes
- 250 electric trucks on long motorway origin-destination pairs, one trip each
- Three 350 kW charging sites, 8 plugs each: 2.8 MW per site
- **Charger locations are the object of study.** Everything else is calibration

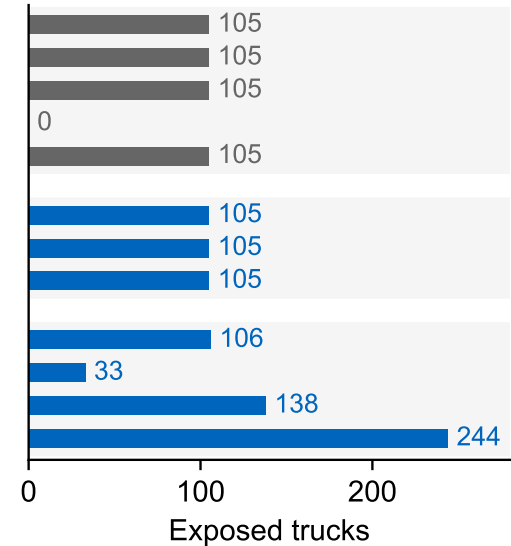
Brandenburg motorway network with its three charging sites

Twelve moves, three families



Rows grouped: v1, v2 geometry, v2 exposure. Dots show each moved site; add shows new-site distance.
Near = nearest other charger (km); geometry window 40–50 km. Exposure is not a campaign measurement.

Design input: p08 baseline
2026-09-13, pre-revision



The twelve perturbations by family, with each moved site's displacement

Range must bind, and plugs must not

	4 plugs	6 plugs	8 plugs
SoC 1.00	1 % stranded wait/service 0.23 fails coverage, range	not run	not run
SoC 0.85	10 % stranded wait/service 0.98 fails coverage, range, stranding	not run	not run
SoC 0.80	13 % stranded wait/service 1.08 fails coverage, queue, range, stranding	6 % stranded wait/service 0.49 fails coverage, range	5 % stranded wait/service 0.17 fails coverage, range
SoC 0.70	30 % stranded wait/service 1.26 fails queue, stranding	14 % stranded wait/service 0.66 fails stranding	4 % stranded wait/service 0.32 passes all four clauses
SoC 0.60	46 % stranded wait/service 0.93 fails coverage, stranding	not run	not run

- If no truck needs to charge, moving a charger changes nothing and there is no signal to measure
- The screen has four clauses: coverage, stranding under 10 %, a queue that is present but not pinned, and range binding
- Of nine cells, **exactly one passes all four**: initial charge 0.70 with 8 plugs. That is the working point the campaign still runs

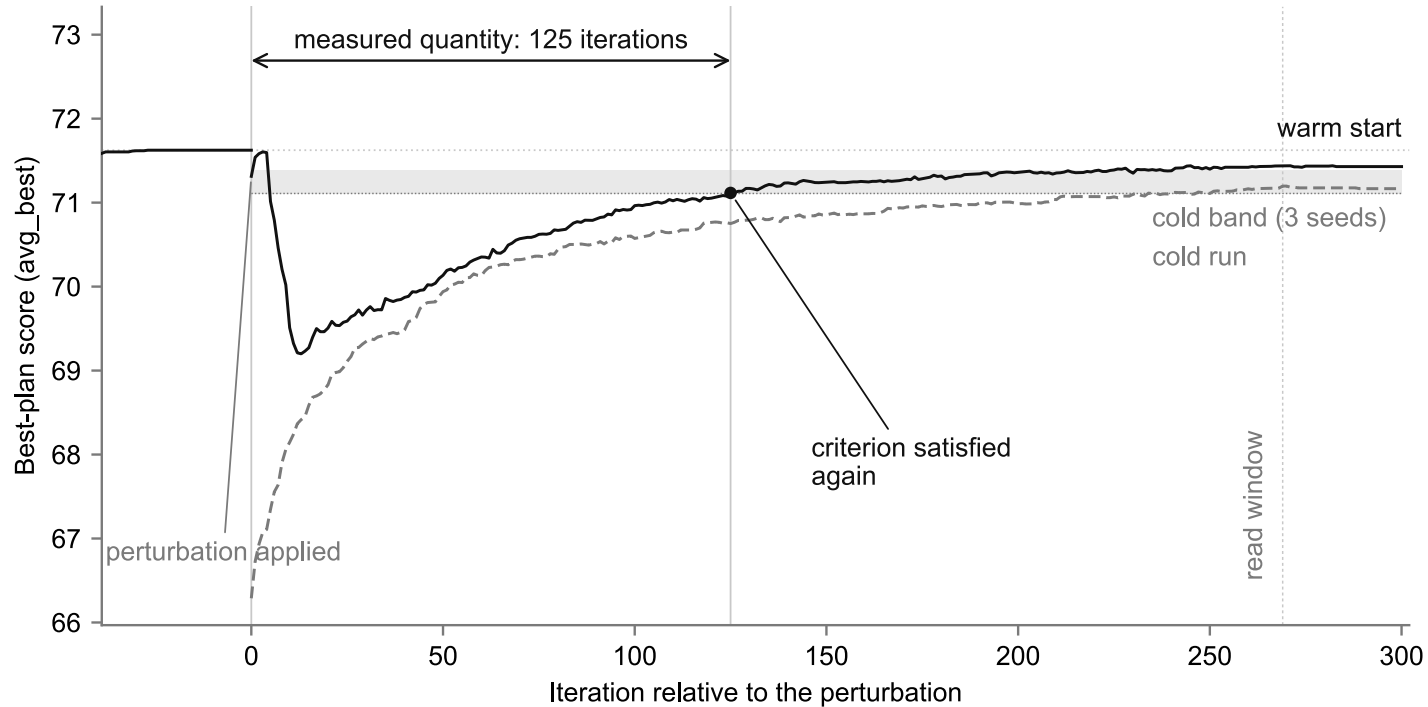
The lower level, revised before the campaign

	Before	Campaign
Charging dwell	Fixed 1 hour	Remaining need + 0.15 capacity reserve Capped at full, configured time floor, queue extends stop
Route randomness	3.0	0
Battery capacity	265 kWh	265 kWh
Initial SoC	0.70	0.70
Plugs per charger	8	8
Replanning weights	ReRoute 0.2 ChangeExpBeta 0.8	ReRoute 0.2 ChangeExpBeta 0.8
Plan memory	5 plans	5 plans

- **Charging dwell:** from a fixed one hour to what the rest of the day needs plus a 0.15 reserve, and queuing now extends the stop
- **Route choice:** randomness $3 \rightarrow 0$, so a difference between two runs is the perturbation and not the router
- **Energy shortfall is priced linearly, 0.20 utils/kWh**, because a flat charge paid trucks to skip charging

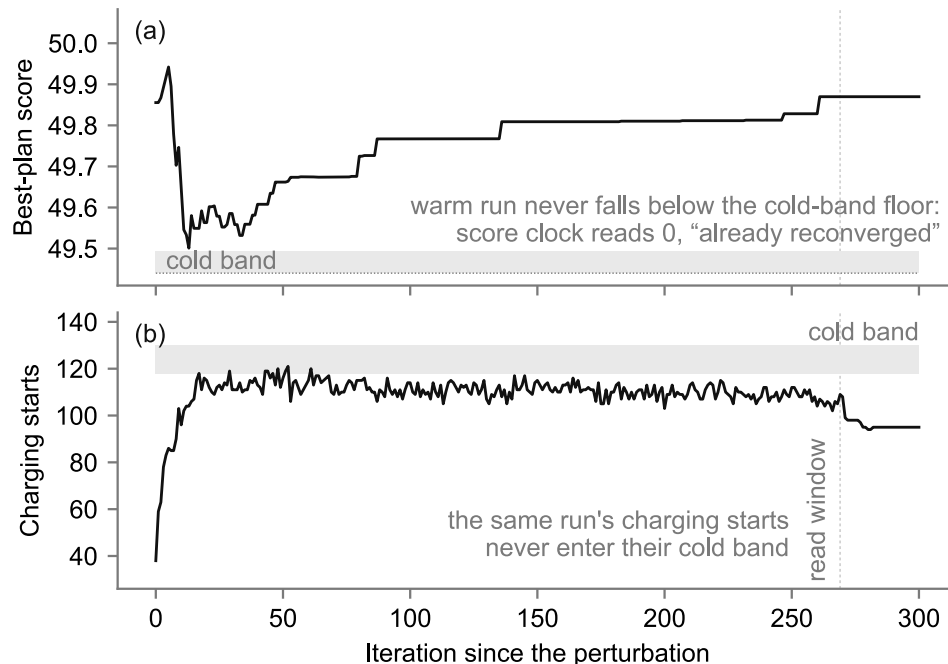
Measuring recovery

What a clock measured



One warm trace after a perturbation against its layout's own cold band, with the clock marked

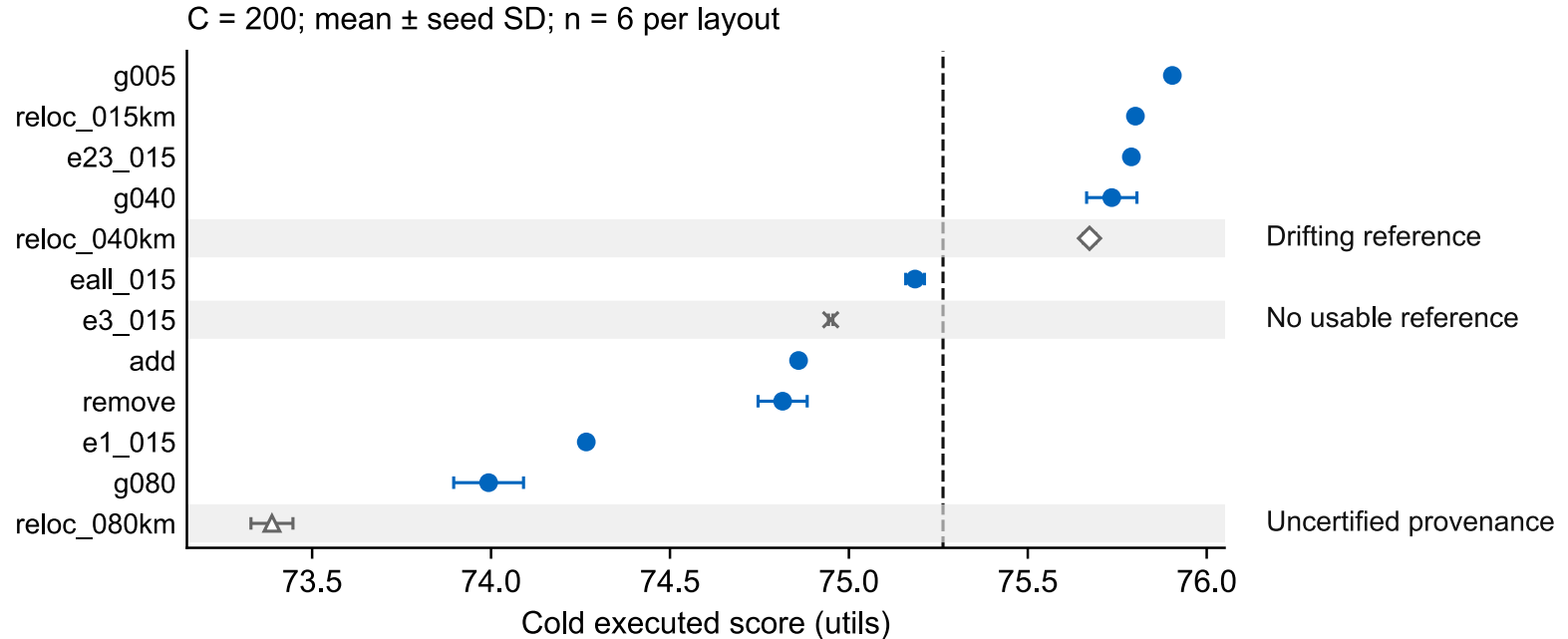
A clock can report recovery that never happened



- The obvious measure is a **clock**: how many iterations until the warm run is back inside its cold band
- A clock returns **0** both for a run that recovered immediately and for one never observed to be damaged at all
- In this earlier-model probe the score reads 0, "already recovered", while the same run's charging starts never re-enter their cold band
- So we do not clock. **We read at a chosen k and test the difference**

A run whose score never falls below the cold floor while its charging starts never re-enter their band

Every layout is scored against its own cold solve

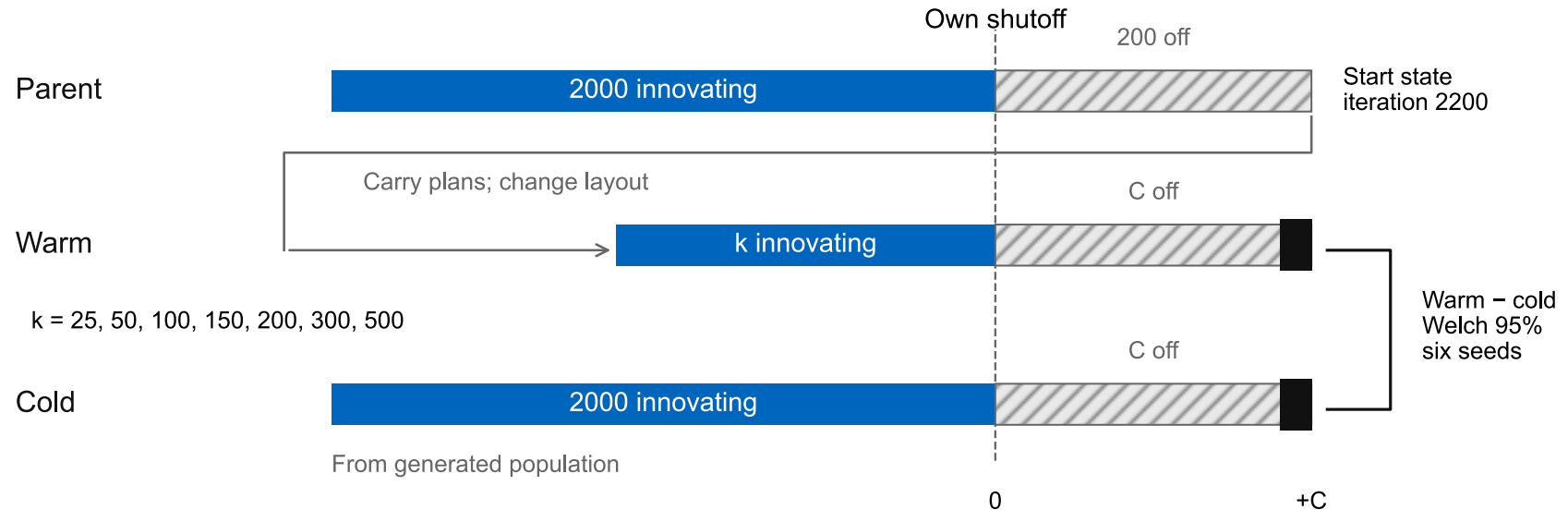


Dashed line: unperturbed six-seed mean, 75.2631 utils (same read window).

Flagged markers show the recorded cold values, not certified usable references.

Twelve treatments' cold executed score with seed spread, against the unperturbed mean, three flagged

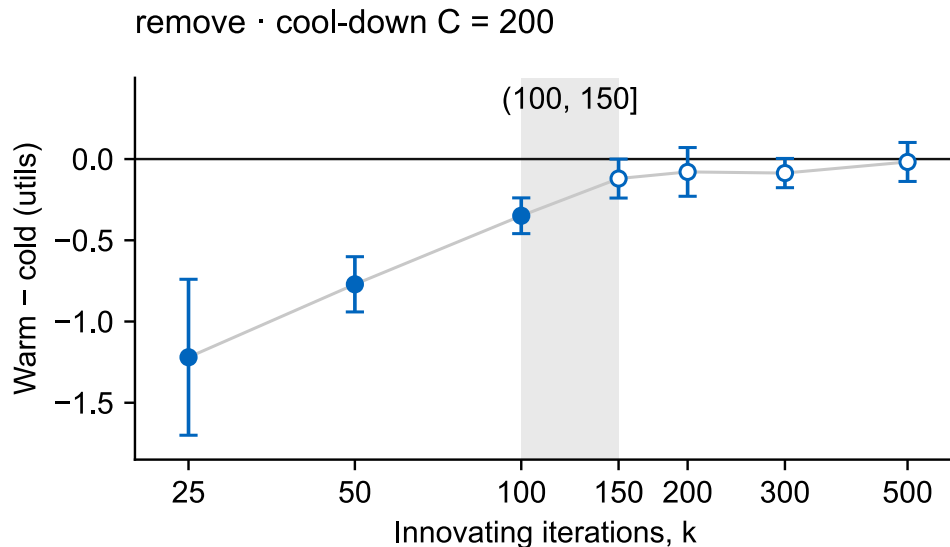
The read: innovate, cool down, compare at the same offset



Blue: innovation on. Hatched: off. Black: trailing 20 iterations ending C after each arm's shutoff.
Common shutoff-relative axis; on-stretches schematic. Campaign C = 200; g040 also has $k = 75$.

Warm arms innovating for k , cooling down for C , read in a trailing window at the same shutoff-relative offset as the cold arm

One treatment, read at every budget



Bars: Welch 95%, six seeds. Filled: resolvably worse.
Open: not resolvably different. Shading: recovery bracket.

Remove: warm minus cold executed score at each k with 95 percent intervals, the bracket shaded

- `remove` takes a charger away. At $k = 25$ its warm arm is 1.22 ± 0.48 utils worse than its own cold arm
- By $k = 500$ the gap is 0.018 ± 0.12 utils: no longer distinguishable from zero
- The deficit resolves between $k = 100$ and $k = 150$, at a cool-down of $C = 200$
- Shorten the cool-down to $C = 100$ and the same treatment brackets at $(150, 200]$. The cool-down is part of the answer

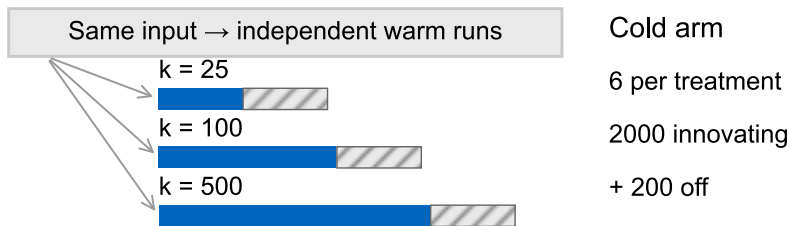
What the campaign found

The campaign

582 comparison runs

One cold parent → one hashed plans file

rb_d_b2000_s4711 · iteration 2200



12 treatments × 6 seeds; k = 25, 50, 100, 150, 200, 300, 500

g040 also k = 75: 510 warm; 72 cold

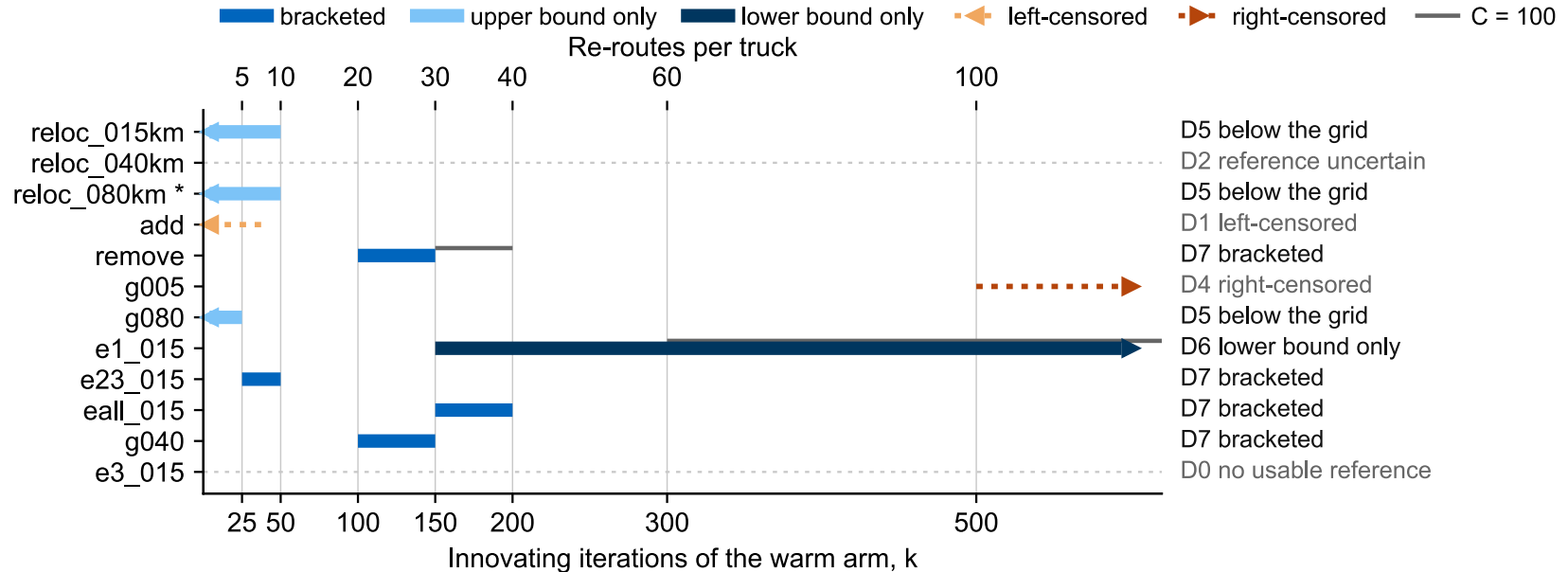
Blue: on; hatched: 200 off. Lengths schematic.

Not bit-reproducible: 109/402 prefix pairs differ.
Every difference compared and reported; runs not nested.
reloc_080km: provenance uncertified.

One parent run, its plans file handed to independent warm runs at each budget, and the cold arm beside them

- One converged parent run, one plans file, handed to every warm run as its starting point
- **Twelve treatments × six seeds × seven budgets** $k \in$, and a cold arm of six runs for each treatment: **582 runs**, 510 warm and 72 cold
- The runs share an input; they are **not nested**. Each is its own run from the same start state
- The build is **not bit-reproducible**: **109 of 402** compared prefixes differ, every difference is reported, and one treatment is uncertified on provenance

Four moves bracket; three show no detectable deficit



* reloc_080km is reported uncensored: five prefix comparisons halt under ADR-075.

Recovery as an interval for every treatment, with censored rows kept on the chart and the shorter cool-down drawn where it moves the bound

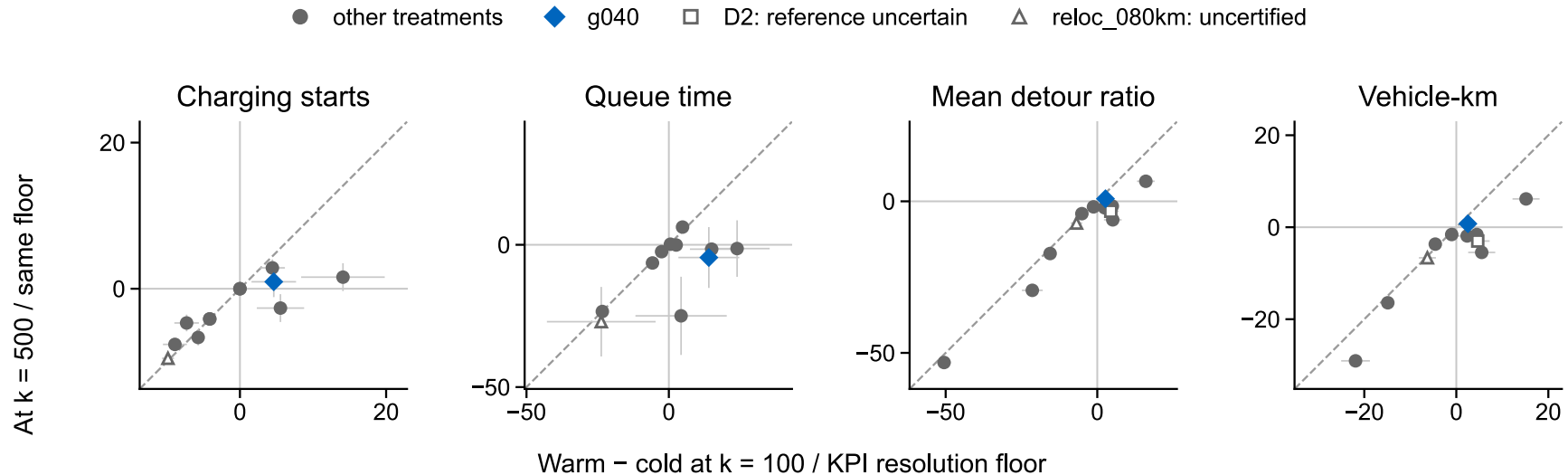
The other five, each for a stated reason

Executed score · cool-down C = 200

add	left-censored	Never left the cold band
g005	right-censored	Still off at k = 500
e1_015	lower bound only	Off at k = 150; no upper bound
reloc_040km	drifting reference	Drift above size, statistically unresolved
e3_015	no usable reference	Cold reference fails flatness

The five treatments that do not bracket, each with its reason in plain words

Score recovery can coexist with physical differences



C = 200. Diagonal: unchanged difference; toward the horizontal zero line: smaller difference. Bars: Welch 95%.
 Scale: each KPI's published resolution floor, not a verdict threshold. Two D0 rows omitted; coincident points overlap.
 Persistence to k = 500 cannot establish that a difference will never close.

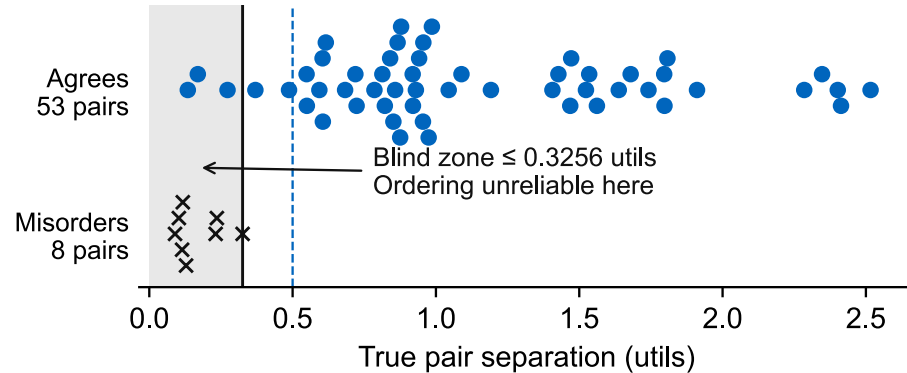
Each treatment's KPI differences at k = 100 and k = 500, scaled by the KPI's resolution floor

What the read is good enough
for

The cheap read gets the order right

$k = 100$, cool-down $C = 100$, weight 0.2

48/48 agree when truth separates by ≥ 0.5 utils

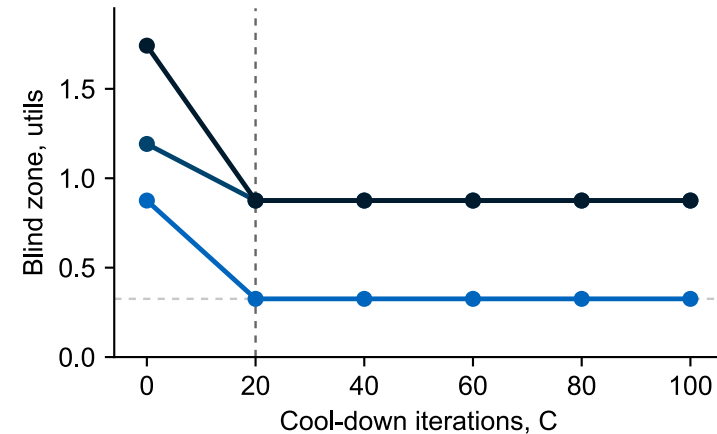
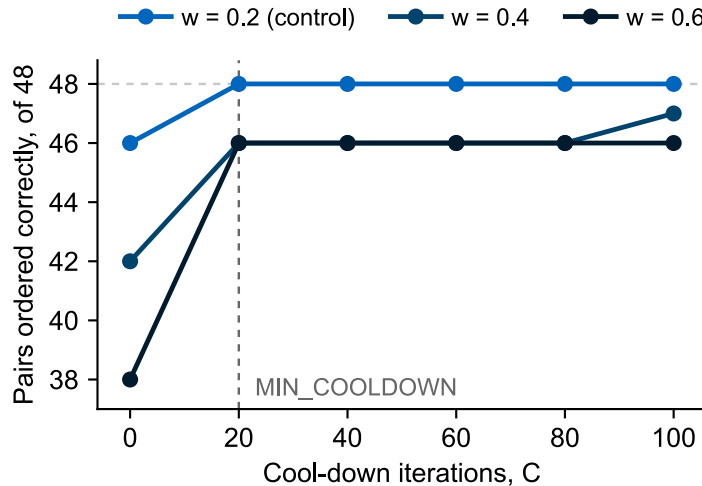


Cold truth: $C = 200$; six seeds in both arms.
 One dot per gradable pair; 5 unresolved pairs omitted.
 All 12 layouts, including flagged references.
 20 three-seed subsets: 48/48 throughout; total 52–53/61.

Every gradable pair of layouts on a separation axis, with the misordered ones marked inside the blind zone

- The loop never asks what a layout is worth. It asks **which layout is best**
- Read at $k = 100$ with a 100-iteration cool-down, **53 of 61** gradable pairs come out in the order their cold solves put them
- **48 of 48** pairs separated by at least half a util: every one correct
- The eight mistakes are all **within 0.33 utils**, among layouts the search would be near-indifferent between. **In this candidate set:** twelve layouts, one start state

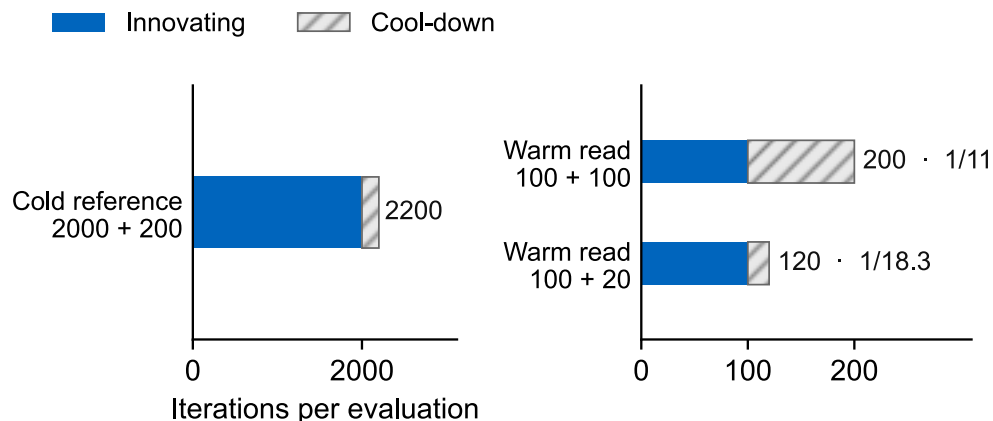
The cool-down does its work in twenty iterations



Budget 20. C = 0 is the within-run read, no cool-down paid. C = 20 is MIN_COOLDOWN, the shortest the read machinery admits, so the floor here is a limit of the grid and not a measured optimum. Weights 0.4 and 0.6 are admitted by no accepted decision (PRN-006): evidence, not a result.

Pairs ordered correctly and the blind zone against cool-down length, for three arms, from no cool-down to C = 100

The price, now that it is earned



$C = 20$ is 38.8% cheaper: 749 against 1223 core-s.
 Same certified runs, read at an earlier offset; no new simulation.
 Correctly ordered pairs fall 53 to 52. Pairs separated by
 ≥ 0.5 utils stay 48/48; blind zone unchanged at 0.3256.

- A cold evaluation is **2200 iterations**. The read we have been quoting, $k = 100$ with $C = 100$, is **200**: about one eleventh
- Take the cool-down down to $C = 20$ and the same read is **120 iterations**: about **one eighteenth**
- Same runs, read at an earlier offset. No new simulation, no new provenance
- The criterion is **ordering well-separated pairs**: those stay at 48 of 48 and the blind zone is unchanged, while one near-tie pair is lost, 53 to 52

*Iterations per evaluation: a cold solve at 2200, the read at $C = 100$ at 200,
 and the same read at $C = 20$ at 120*

Where this leaves us

What this buys

Warm-starting a converged run evaluates a candidate charger layout well enough to rank it, in the simplest setting we could build: one converged state, one move at a time, twelve directions.

- A cold evaluation is **2200 iterations**. The read that ranks, $k = 100$ cooled for 20, is **120**: about **one eighteenth**
- So a search over a hundred candidate layouts costs about **five and a half cold solves**, not a hundred. That is the difference between a loop that runs and one that does not
- Earned on **this candidate set**: twelve layouts, one start state, charger locations only. Prices, technology shares and capacity were not perturbed

From here

- **A second starting state.** Every number here descends from one converged parent. Nothing yet separates a property of the method from a property of that parent
- **Layouts a search proposes.** The twelve were chosen to span move sizes, not drawn from what an optimiser would query. The blind zone was measured on them
- **Chained moves, warm from warm.** The loop never evaluates one move from the parent; it evaluates a path. Whether the error compounds along it is unmeasured
- **A stopping rule that lives inside the loop.** Today the loop runs on the fixed budget. A rule calibrated on the parent failed on replay: every layout has its own settled noise
- **What makes one layout close to another.** Neither kilometres moved nor trucks affected predicted the cost of warming one layout from another. Without a measure that does, the loop cannot choose which evaluated layout to warm from, or how long to run

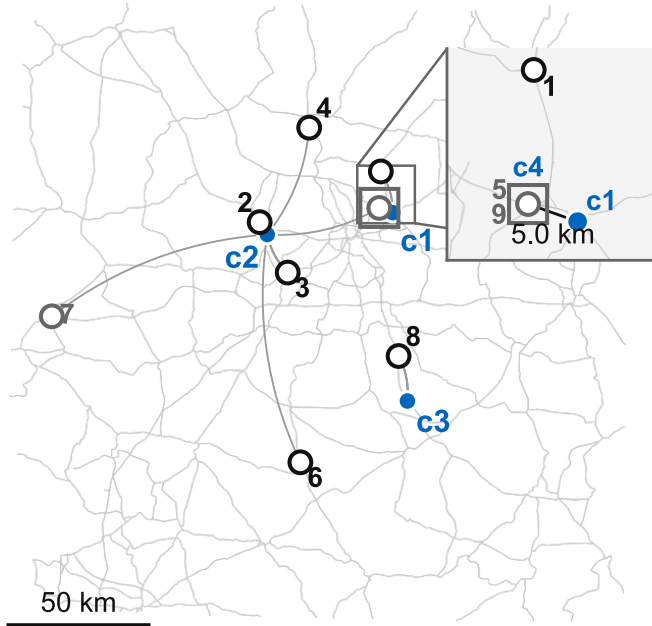
Shahriar Iqbal Zame
shahriar.zame@tum.de

Technical University of Munich
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Paris, 28 September 2026



Backup

Where the moves are on the network



- 1 c1 +14.9 km: e1_015, eall_015
- 2 c2 +5.0 km: g005
- 3 c2 +15.0 km: e23_015, eall_015, reloc_015km
- 4 c2 +39.9 km: g040
- 5 c2 +40.0 km: reloc_040km
- 6 c2 +80.0 km: g080
- 7 c2 +80.0 km: reloc_080km
- 8 c3 +15.9 km: e23_015, e3_015, eall_015
- 9 c4 added: add

black: v2, nearest other charger held to 40 to 50 km (ADR-006)
 grey: v1, unconstrained; reloc_040km landed 5 km from c1 (OI-004)
 square: c4, the charger add gains. It lands on reloc_040km's destination link, so it sits 5 km from c1 too. Inset: the c1 neighbourhood, 20 km across

The perturbation layouts drawn on the Brandenburg network, with the charger-1 neighbourhood zoomed

Every treatment and KPI, with its disposition

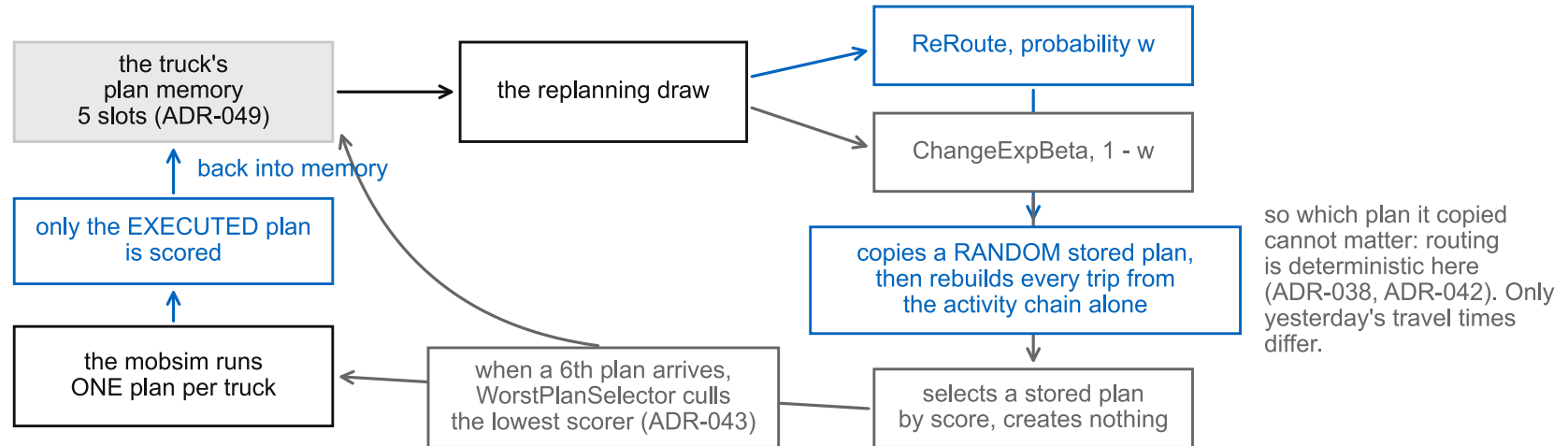
	executed score	charging starts	queue time	mean detour ratio	vkt	energy charged	best score
reloc_015km	D5	D4	D3	D4	D4	D4	D5
reloc_040km	D2	D7	D4	D4	D4	D4	D2
reloc_080km *	D5	D4	D3	D4	D4	D4	D0
add	D1	D5	D4	D4	D0	D4	D0
remove	D7	D3	D7	D4	D4	D3	D0
g005	D4	D4	D3	D4	D4	D4	D4
g080	D5	D4	D3	D3	D3	D3	D0
e1_015	D6	D4	D0	D3	D3	D4	D4
e23_015	D7	D4	D3	D3	D3	D4	D5
eall_015	D7	D4	D7	D2	D2	D4	D7
g040	D7	D3	D7	D7	D7	D6	D0
e3_015	D0	D5	D4	D4	D4	D4	D1

■ D7 bracketed
■ D5 below the grid
■ D1 left-censored
■ D3 not monotone
■ D6 lower bound only
■ D4 right-censored
■ D2 reference uncertain
■ D0 no usable reference

* reloc_080km is reported uncertified: five prefix comparisons halt under ADR-075.

The disposition of all 84 treatment-by-KPI rows at $C = 200$

One MATSim iteration, in full



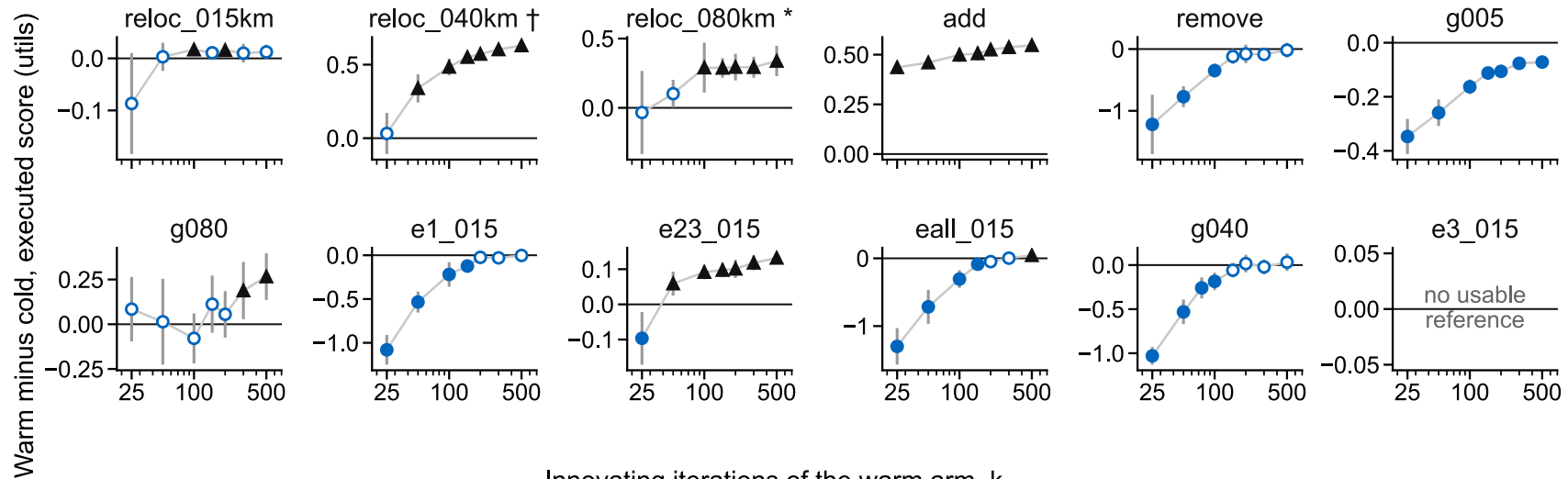
A cull is irreversible, and a plan is culled on however many scores it has collected so far. That is why the same budget spent at a heavier weight buys a worse answer (F41).

Verified against the jar; see experiments/reconvergence/docs/matsim-internals.md sections 1, 2, 4 and 7. No numbers to assert.

One MATSim iteration: replanning, mobsim, scoring, and the five-plan memory

Seven moves score above their cold reference

● resolvably worse than cold ○ not resolvably different ▲ resolvably above cold



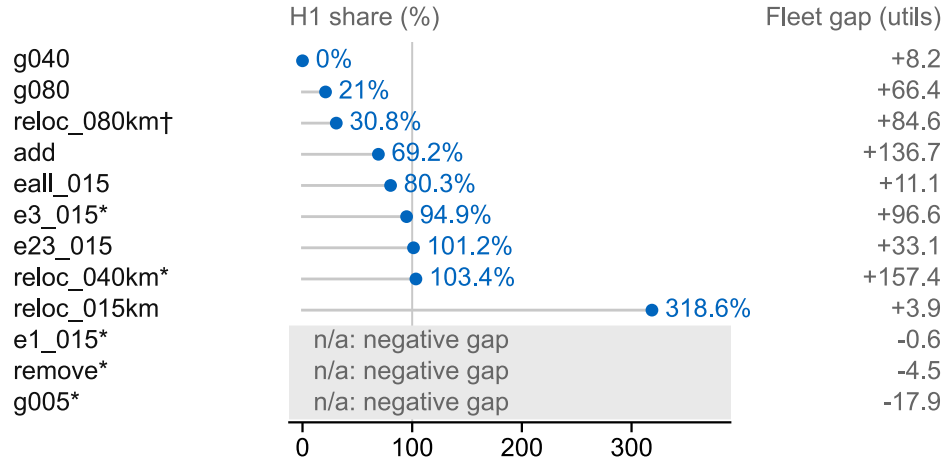
* reloc_080km is reported uncertified: five prefix comparisons halt under ADR-075.

† reloc_040km's cold reference drifts, so "above" there is measured against a target that does not sit still.

Warm minus cold executed score against k for every treatment, with intervals, above zero not clipped

What a higher score can come from

H1 share varies across treatments

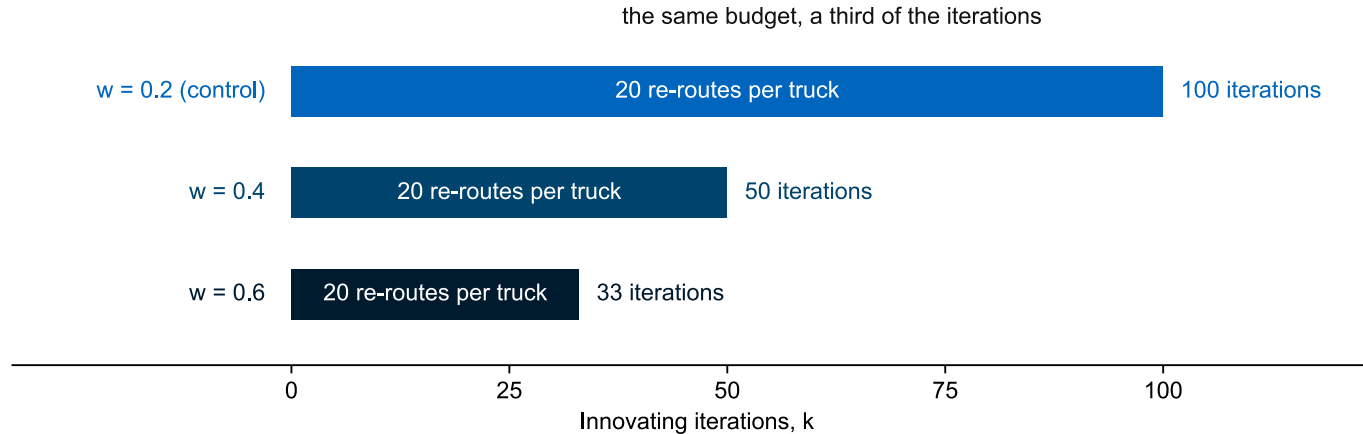


H1: inferred stored-score signature; not isolated.
 >100%: other trucks offset gains. * Pattern only.
 † Provenance uncertified. Shortfall can affect scores.
 No persistence claim beyond compared prefixes.

- Some trucks keep a plan inherited from the unperturbed layout, one the perturbed layout's router never writes
- On **add**, those carriers account for **94.6** of a **136.7-util fleet gap**, about **69 %**
- But the share is **uneven**: 0 % on **g040**, about 21 % on **g080**, and above 100 % on others, where the rest of the fleet moved the other way
- So this is a mechanism, not **the** mechanism, and it is inferred rather than isolated

Fleet gap and the inherited-plan carriers' share of it, per treatment, with the negative-gap rows excluded

What we are trading

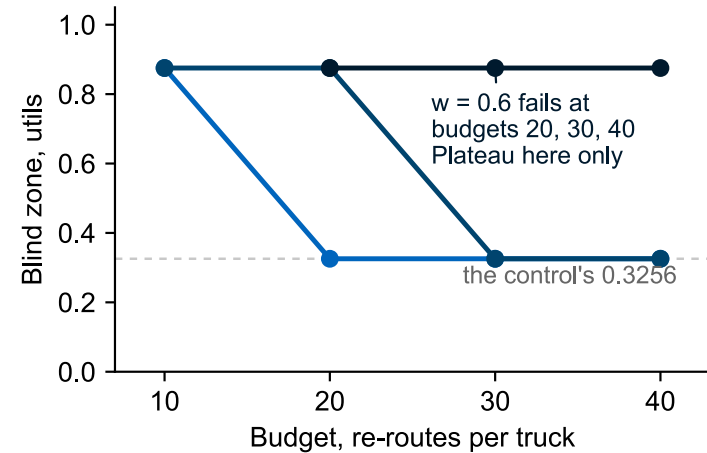
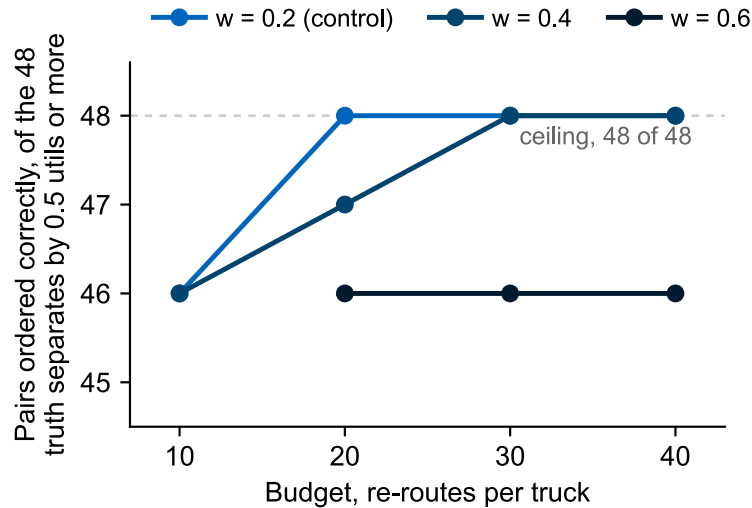


Budget is the matching unit (ADR-031): $k \times$ the replanning weight, the mean number of times a truck is re-routed.

An iteration costs about the same at every weight, so fewer iterations is the whole saving. Weights 0.4 and 0.6 are admitted by no accepted decision (PRN-006): evidence, not a result.

Twenty re-routes per truck is 100 iterations at weight 0.2, 50 at 0.4 and 33 at 0.6

Re-routing harder does not buy the same answer

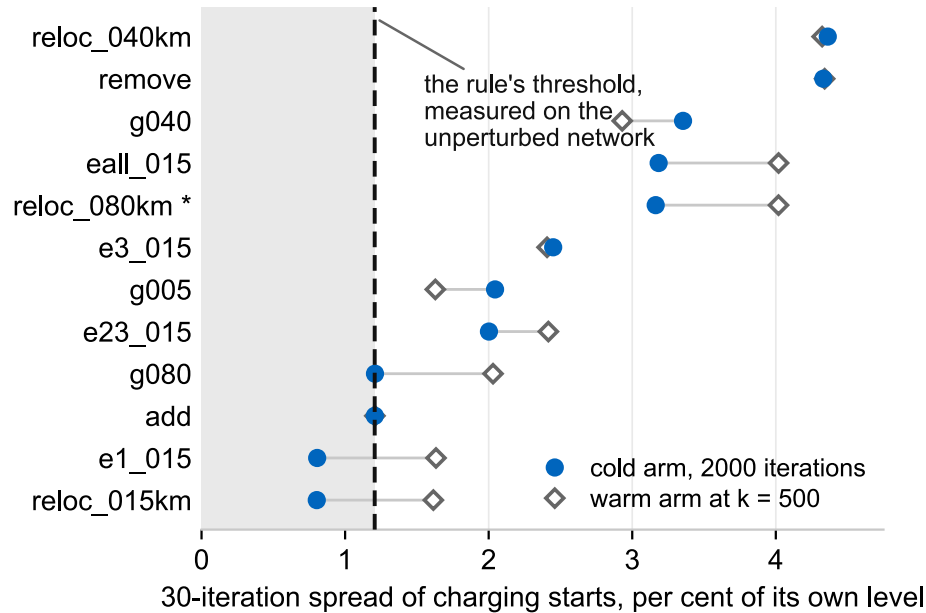


Experimental schedule comparison; ladder-wave provenance not certified.

Cooled read, $C = 100$; no extrapolation beyond tested budgets. Weights 0.4 and 0.6 are admitted by no accepted decision (PRN-006): evidence, not a result.

The three arms against budget: the control at its ceiling, weight 0.4 arriving late, weight 0.6 flat

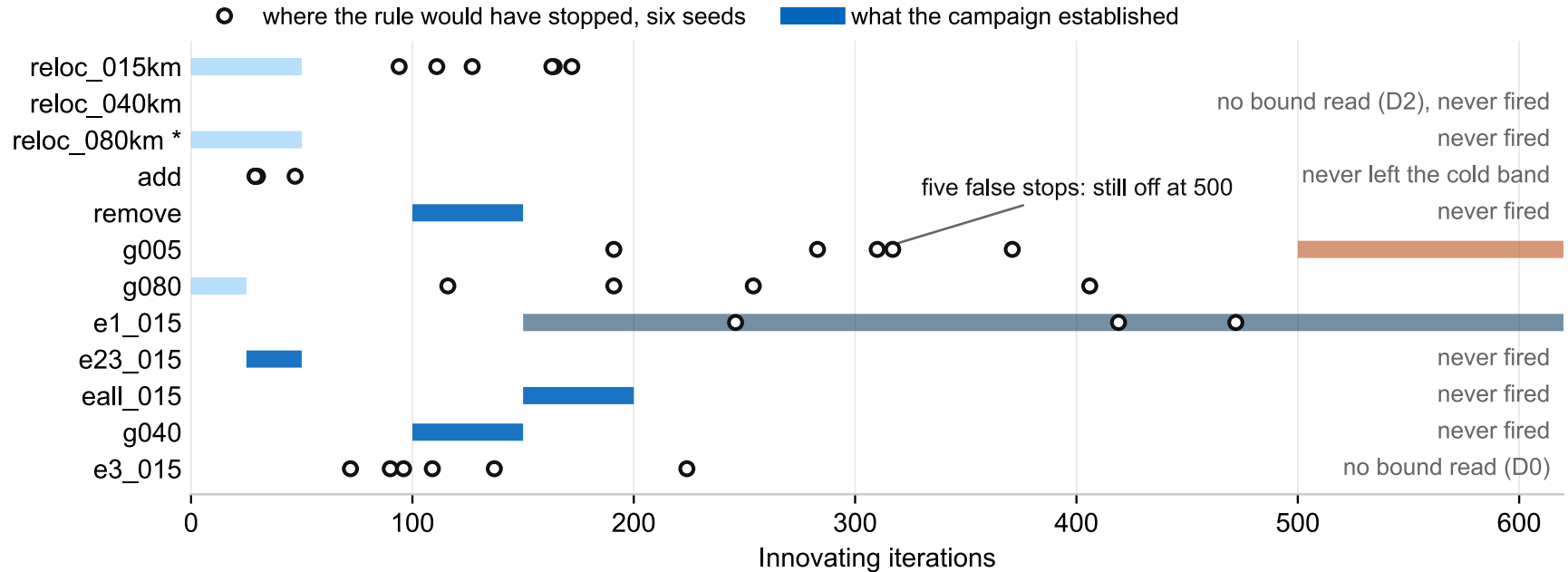
Inside the loop there is no cold arm



- Every number so far used a cold arm. The loop has none, so it must stop a run by watching that run alone
- Our rule asks each layout to hold as steady as the **unperturbed network**. Eight of twelve are **1.7 to 3.6 times noisier**, after 2000 iterations of their own
- On **remove** and **reloc_040km** the warm run is already as steady as its layout ever gets, and the gate is still **3.6 times too tight**
- Steadiness is not recovery. The loop runs today on Act 5's **fixed budget**; a within-run rule must take its scale from the run itself

Per layout, the settled spread of charging starts in its cold arm and in its warm arm at $k = 500$, against the threshold calibrated on the unperturbed network

Where the rule would have stopped



Per layout, the campaign's own recovery bound as a bar and the six firing iterations of the replayed stopping rule as points